QueMore

Software Engineering & Solutions

SOFTWARE REQUIREMENTS SPECIFICATION

DesignArc

Furniture Manufacturing Workflow Platform — Web & Mobile

Conforms to ISO/IEC/IEEE 29148:2018 — Requirements Engineering

Revision 1.1 · 18 July 2026 · Status: Draft for Review

Document Control	
Document Title	Software Requirements Specification — DesignArc
System Name	DesignArc Furniture Manufacturing Workflow Platform
Client / Business Owner	DesignArc — Interior Designers & Consultants
Prepared By	QueMore — Business Analysis Team
Governing Standard	ISO/IEC/IEEE 29148:2018 (supersedes IEEE 830-1998)
Revision	1.1
Status	Draft for stakeholder review
Date	18 July 2026

Revision History

Version	Date	Author	Summary of change
0.1	10 Jul 2026	QueMore BA Team	Initial draft from discovery workshop
1.0	18 Jul 2026	QueMore BA Team	First baseline candidate for stakeholder review
1.1	18 Jul 2026	QueMore BA Team	Revised pricing flow: price is set after Supervisor approval; worker then accepts or declines

> **QueMore requirements philosophy**
> A requirement that cannot be tested does not exist. A requirement that can be interpreted two ways will be built the wrong way. Every requirement in this document is written to be necessary, unambiguous, feasible, verifiable, traceable, and complete.

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1 Introduction

This Software Requirements Specification (SRS) establishes the shared understanding between DesignArc and QueMore of the system to be built. Sections 1 and 2 set the context; Section 3 defines every requirement the development team will build and the QA team will verify.

1.1 Purpose

This SRS defines the functional and non-functional requirements for the DesignArc Furniture Manufacturing Workflow Platform. It establishes a verifiable agreement between DesignArc (Business Owner) and QueMore (Development Team) on what the system shall do. This SRS forms the baseline for solution design, development, verification, acceptance testing, and future change control. It is intended for the DesignArc business owner and supervisors, and for QueMore's design, development, and QA teams.

1.2 Scope

DesignArc is a furniture-manufacturing and interior-design business. Work is organised into projects, each broken into job cards that pass through a carpentry stage and a painting stage. The platform digitises the assignment, execution, inspection, post-inspection pricing, payment, and reporting of that work across a web application (for the Admin) and a mobile application (for carpenters, painters, and supervisors).

In Scope — MVP

- User accounts and role-based access for Admin, Carpenter, Painter, and Supervisor.
- Project creation and management by the Admin.
- Job cards with Carpentry and Painting stages and design attachments (no price is set at creation).
- Work execution: worker marks a stage 'Ready for Inspection'; the Supervisor inspects, uploads photos, and approves or rejects (rework).
- Post-inspection pricing: after the Supervisor approves a stage, the Admin sets the price; the worker then accepts or declines it, with a full pricing history.
- In-app notifications on every workflow handoff.
- Per-stage earning records with Paid / Unpaid status.
- A basic Admin dashboard.

In Scope — Version 1.0

- Full Admin dashboard with KPIs and an approval action queue.
- Monthly earnings reports per worker, exportable as PDF.
- Push notifications to mobile devices.
- Project completion roll-up and worker workload views.

Out of Scope

- Online payment disbursement or payroll/bank integration (the system records payment status only).
- Materials, stock, or inventory management.
- A client-facing portal.
- Integration with external accounting systems.
- Multi-language user interface and full offline-first operation.

Business Benefits

Benefit	Measurable outcome
Eliminate paper/phone job tracking	All job assignments and statuses captured digitally, no verbal handoffs
Transparent pricing	Price is set after work is approved; every proposed and revised price is recorded with the worker's accept/decline
Proof of work	Photographic evidence attached to every approved stage
Accurate worker payments	Per-stage earnings computed automatically from agreed amounts
Real-time visibility	Live project completion and workforce status on the Admin dashboard

1.3 Product Perspective

The platform is a new, self-contained system comprising a responsive web application for the Admin and a cross-platform mobile application for field roles, served by a shared backend API and database. It does not replace an existing system. The sub-sections below follow the IEEE 29148 clause structure; every sub-section is completed, and 'none' means the item was considered and ruled out.

Sub-section	Description
1.3.1 System Interfaces	None required for MVP. Future accounting/payroll integration is deferred (see Out of Scope).
1.3.2 User Interfaces	Responsive web (Admin) and native-style mobile (field roles). UI follows the DesignArc brand identity — forest green (#4E7A54) primary, brick red (#B23A2E) accent, cream (#F7F4EE) background.
1.3.3 Hardware Interfaces	Mobile device camera for capturing inspection photographs. Otherwise standard internet-connected devices.
1.3.4 Software Interfaces	Backend REST API (JSON payloads), a push-notification service, and cloud file/object storage for design files and photos.
1.3.5 Communications	All traffic over HTTPS/TLS 1.2+. Authentication via signed tokens (JWT). Timestamps in ISO 8601. Payloads in JSON.
1.3.6 Memory Constraints	Inspection photo size capped (see TBD-05). Server storage sized for design files and photo history; thresholds drive NFR-P targets.
1.3.7 Operations	Automated daily database backups, health monitoring and alerting, and a defined maintenance window; owned by QueMore Operations.
1.3.8 Site Adaptation	Single-tenant deployment for DesignArc. Currency/locale configuration (see TBD-02).
1.3.9 Services	Cloud hosting, object storage, and push-notification service required across dev, staging, UAT, and production.

1.4 Product Functions

The following is the high-level summary of what the system does. Detailed 'shall' statements live in Section 3.2.

Function Group	Major Capabilities
Access & Roles	Authentication, role-based access control, user account management
Project Management	Create/edit/archive projects, project completion roll-up

Function Group	Major Capabilities
Job Cards & Stages	Create job cards, define carpentry/painting stages, attach designs, track status
Stage Assignment	Assign each stage to a carpenter or painter with the design brief (no price)
Work & Inspection	Mark ready, Supervisor photo upload, approve or reject for rework
Post-Inspection Pricing	After approval the Admin sets the price; worker accepts or declines; pricing history
Notifications	In-app and push alerts on every workflow handoff
Payments	Per-stage earning records, Paid/Unpaid status, mark as paid
Dashboard & Reports	Admin KPI dashboard, approval queue, worker monthly earnings reports

1.5 User Characteristics

Role	Description	Proficiency	Key Characteristics
Admin	Creates projects and job cards, approves amounts, manages payments, monitors progress	Intermediate	Desktop/web, office setting, daily use
Carpenter	Executes carpentry stages; marks work ready; accepts or declines the price set after approval	Beginner–Intermediate	Mobile, workshop floor, intermittent connectivity, daily
Painter	Executes painting stages; same workflow as Carpenter	Beginner–Intermediate	Mobile, workshop floor, daily
Supervisor	Inspects finished stages, uploads photos, approves or rejects work	Intermediate	Mobile with camera, on the floor, multiple times per day

1.6 Limitations (Constraints)

1.6.1 Regulatory

The system shall handle personal data (user identities, payment records) in line with applicable data-protection obligations, including the right to have personal data corrected or erased where no legal retention requirement applies.

1.6.2 Design and Implementation

- Passwords shall be stored using a strong adaptive hash (bcrypt, minimum cost factor 12).
- All network communication shall use HTTPS/TLS 1.2 or higher.
- The mobile application shall run on current supported versions of Android and iOS.
- The web application shall support current versions of Chrome, Edge, Firefox, and Safari.

1.6.3 Criticality

Business-critical functions are inspection sign-off, price setting and acceptance, and payment records, because they determine what workers are paid. Loss or corruption of pricing history, inspection photos, or earning records is not acceptable; these must be recoverable from backup. Notification delivery is important but non-critical — a missed notification must not block the underlying workflow.

1.7 Assumptions, Dependencies & Deferred Requirements

1.7.1 Assumptions

ID	Assumption	Risk if incorrect
ASM-01	All field workers have a smartphone capable of running the mobile app	Field workflow becomes unusable; paper fallback required
ASM-02	Each worker has a unique identifying account (email or phone)	Jobs and payments cannot be reliably attributed
ASM-03	Reliable internet is available at the workshop for photo uploads	Inspection sign-off is delayed
ASM-04	One Supervisor role inspects both carpentry and painting	Workflow must be redesigned for separate inspectors
ASM-05	The app records payments but does not disburse money	Scope expands significantly if disbursement is expected

1.7.2 Dependencies

DesignArc must provide brand assets and confirm the final role definitions before UI design begins. DesignArc must supply the initial list of users and their roles before User Acceptance Testing. QueMore must provision cloud hosting, object storage, and push-notification credentials before the staging environment is stood up.

1.7.3 Deferred Requirements

Feature	Target version	Rationale
Separate Finance role for payments	v2.0	Confirm whether Admin retains payment authority (TBD-03)
Materials / inventory tracking	v3.0	Not required for core workflow value
Client-facing portal	v3.0	Depends on core platform being stable first
Accounting system integration	v3.0	Requires confirmed accounting package

1.8 Definitions

Term	Definition
Project	A top-level unit of work for a DesignArc client, containing one or more job cards
Job Card	A single item of furniture work within a project, made up of one or more stages
Stage	A phase of a job card assigned to one worker; either a Carpentry stage or a Painting stage
Accepted Price	The price for a stage after the worker accepts it, which follows Supervisor approval; the basis for the worker's payment
Pricing History	The append-only record of every price proposed, revised, accepted, and declined for a stage
Ready for Inspection	A stage status set by the worker to signal that work is complete and awaiting Supervisor inspection

Term	Definition
Approved	A stage status set by the Supervisor after inspection, confirming the work is accepted; pricing begins only after this
Completed	A stage status reached once the worker has accepted the price set after approval
Rework	The return of a rejected stage to the same worker for correction
Earning Entry	A per-stage record of what a worker has earned once the stage price is Accepted
Monthly Report	A per-worker summary of jobs, earnings, and payment status for a selected calendar month

1.9 Acronyms & Abbreviations

Acronym	Meaning
API	Application Programming Interface
JWT	JSON Web Token
MVP	Minimum Viable Product
NFR	Non-Functional Requirement
PDF	Portable Document Format
RBAC	Role-Based Access Control
RTM	Requirement Traceability Matrix
SRS	Software Requirements Specification
TBD	To Be Determined
TLS	Transport Layer Security
UAT	User Acceptance Testing
UI / UX	User Interface / User Experience
WCAG	Web Content Accessibility Guidelines

2 References

Ref.	Document	Source	Date
[1]	ISO/IEC/IEEE 29148:2018 — Systems and software engineering — Requirements engineering	ISO / IEC / IEEE	2018
[2]	DesignArc workflow discovery notes (project → job card → carpentry → painting → inspection → payment)	DesignArc / QueMore workshop	Jul 2026
[3]	DesignArc brand identity (logo and colour palette)	DesignArc	2026
[4]	OWASP Application Security Verification Standard (ASVS)	OWASP	Current
[5]	Web Content Accessibility Guidelines (WCAG) 2.1 AA	W3C	2018

3 Specific Requirements

This section is the core of the SRS. Every requirement uses the word 'shall' to denote a mandatory behaviour, and each is written to be individually testable.

3.1 External Interfaces

The system boundary is defined by the following overarching interface rules. Per-feature interface behaviour appears within each function block in Section 3.2.

- The mobile app shall access the device camera to capture inspection photographs, subject to the user's device permission.
- The system shall exchange data between clients and server exclusively over an authenticated HTTPS/TLS REST API using JSON payloads.
- The system shall store design files and inspection photos in cloud object storage and serve them only to authenticated, authorised users.
- The system shall send push notifications to registered mobile devices through the platform push-notification service.

3.2 Functions (Functional Requirements)

Requirement writing formula
The system shall [perform action] [under condition] [with measurable constraint]. Each requirement expresses exactly one behaviour — no hidden 'and' — so that it maps to a single test case.

3.2.1 User Authentication & Role Management

Controls who can access the platform and what each role may do. Four roles exist: Admin, Carpenter, Painter, and Supervisor. The Admin provisions accounts; access to every other function depends on the authenticated role.

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
A user submits login credentials	The system authenticates the user and loads their role-specific home screen
An Admin creates a new user and assigns a role	The system provisions the account and notifies the user
A user attempts an action their role does not permit	The system blocks the action and records the attempt
A user requests a password reset	The system emails a verified reset link

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-1.1	Authenticate	High	authenticate users via email/username and password before granting access to any role-specific function.
FR-1.2	Enforce RBAC	High	enforce role-based access control for the four roles so each user can access only the functions permitted to their role.

Req. ID	Title	Priority	The system shall …
FR-1.3	Manage users	High	allow an Admin to create, edit, and deactivate user accounts and assign exactly one primary role to each account.
FR-1.4	Block & log	Medium	reject any action not authorised for a user's role and record the attempt with user ID, timestamp, and attempted action.
FR-1.5	Password reset	Medium	allow a user to reset a forgotten password through a verified email link.

Business Rules

Rule ID	Rule statement
BR-1.1	Each user account shall have exactly one active primary role at any time.
BR-1.2	Only an Admin shall create or deactivate user accounts.

3.2.2 Project Management

Lets the Admin create and manage projects, which group related job cards for a single client engagement.

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
An Admin submits the new-project form	The system creates the project with a unique ID and status 'Active'
An Admin opens a project	The system displays the project details and its completion percentage
An Admin archives a project	The system marks the project archived and hides it from active lists

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-2.1	Create project	High	allow an Admin to create a Project with a name, client, description, and optional deadline.
FR-2.2	Unique ID	High	assign each Project a unique identifier upon creation.
FR-2.3	Completion roll-up	Medium	calculate and display each Project's completion percentage as the proportion of its job-card stages marked Approved.
FR-2.4	Edit / archive	Medium	allow an Admin to edit or archive a Project.
FR-2.5	Protect records	Medium	prevent deletion of a Project that has job cards with recorded payments; such projects shall only be archived.

Business Rules

Rule ID	Rule statement
BR-2.1	A Project shall contain zero or more Job Cards.

Rule ID	Rule statement
BR-2.2	Only an Admin shall create, edit, or archive a Project.

3.2.3 Job Card & Stage Management

A job card is a unit of furniture work within a project. Each job card has one or more sequential stages — Carpentry, then optionally Painting — each with its own assignee, agreed amount, status, and photos. Modelling each stage as its own record keeps the two identical workflows reusable and payments per-worker clean.

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
An Admin adds a job card with design files (no price)	The system creates the job card and its configured stages
A Carpentry stage is Approved	The system unlocks the Painting stage of the same job card
A user opens a job card	The system displays the stage tracker, assignees, amounts, and attached photos

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-3.1	Create job card	High	allow an Admin to create a Job Card under a Project with a title, description, and one or more design attachments; no price is set at this stage.
FR-3.2	Stage model	High	support Job Card stages of type Carpentry and Painting, each with its own assignee, agreed amount, status, inspection photos, and timestamps.
FR-3.3	Configure stages	Medium	allow an Admin to configure which stages apply to a Job Card (Carpentry only, or Carpentry followed by Painting).
FR-3.4	Sequence gate	High	unlock the Painting stage only after the Carpentry stage of the same Job Card is marked Approved.
FR-3.5	Status machine	High	maintain a status for each stage from the set: Assigned, In Progress, Ready for Inspection, Approved, Rejected, Price Proposed, Price Declined, Price Accepted, Completed.
FR-3.6	Store media	Medium	store all design attachments and inspection photos against the Job Card and make them viewable to the Admin, the assigned worker, and the Supervisor.

Business Rules

Rule ID	Rule statement
BR-3.1	A Job Card stage price shall be set only after the stage has been Approved by the Supervisor; no price exists on a stage before then.
BR-3.2	Painting shall not start before Carpentry is Approved by the Supervisor on the same Job Card.

3.2.4 Stage Assignment

Assigns each stage to a worker at the start of the work, with the design brief but no price. Under the revised flow, pricing is deferred until after the Supervisor has approved the finished work (see Section 3.2.6).

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
An Admin assigns a stage to a carpenter or painter with the design brief	The system notifies the worker of the assignment
The worker opens the assigned stage	The system displays the design attachments so work can begin

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-4.1	Assign stage	High	allow an Admin to assign a Job Card stage to a specific Carpenter (Carpentry) or Painter (Painting) together with the design brief and without a price.
FR-4.2	Notify assignee	High	notify the assigned worker when a stage is assigned to them.
FR-4.3	Provide design	Medium	make the design attachments for an assigned stage available to the assigned worker.

Business Rules

Rule ID	Rule statement
BR-4.1	A stage shall be assigned to exactly one worker whose role matches the stage type (Carpenter for Carpentry, Painter for Painting).
BR-4.2	No price shall be recorded on a stage at assignment time.

3.2.5 Work Execution & Supervisor Inspection

The worker performs the assigned work and marks it ready. The Supervisor — not the worker — physically inspects the finished work, uploads the photographs, and approves or rejects it. Supervisor approval is what triggers the pricing step in Section 3.2.6. Placing photo capture and sign-off with the Supervisor gives a clean, independent audit trail.

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
The worker marks an in-progress stage 'Ready for Inspection'	The system sets the status and notifies the Supervisor
The Supervisor inspects, uploads photos, and approves	The system sets the stage to Approved and records photos, notes, identity, and time
The Supervisor rejects with a reason	The system returns the stage to In Progress and notifies the worker for rework

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-5.1	Mark ready	High	allow the assigned worker to mark a stage as Ready for Inspection once work has commenced (no price is required at this point).
FR-5.2	Notify Supervisor	High	notify the Supervisor when any stage is marked Ready for Inspection.
FR-5.3	Supervisor photos	High	allow only the Supervisor to upload inspection photographs against a stage.
FR-5.4	Approve stage	High	allow the Supervisor to Approve a stage, setting its status to Approved and recording the Supervisor's identity, photos, notes, and timestamp, which makes the stage eligible for pricing.
FR-5.5	Reject / rework	High	allow the Supervisor to Reject a stage with a mandatory reason, returning it to In Progress and notifying the assigned worker.
FR-5.6	Require evidence	Medium	require at least one inspection photograph before a stage can be Approved.

Business Rules

Rule ID	Rule statement
BR-5.1	Only the Supervisor shall upload inspection photos and mark a stage Approved or Rejected.
BR-5.2	A stage rejection shall return the same stage to the same assigned worker, not create a new stage.

3.2.6 Post-Inspection Pricing

Pricing happens only after the Supervisor approves a stage. The Admin sets the price for the approved stage; the assigned worker then Accepts or Declines it. If declined, the Admin sets a revised price and the worker reviews again. Acceptance finalises the stage (status Completed) and creates the worker's earning. This is the key change from the earlier flow, where a price was proposed at job-card creation.

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
A stage is Approved by the Supervisor	The system enables the Admin to set the stage price
The Admin sets the price	The system sets the stage to Price Proposed and notifies the assigned worker
The worker accepts the price	The system sets the stage to Price Accepted, then Completed, and creates an earning entry
The worker declines the price	The system sets the stage to Price Declined and notifies the Admin
The Admin sets a revised price	The system returns the stage to Price Proposed for the worker to review again

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-6.1	Enable pricing	High	enable price entry for a stage only after that stage has been Approved by the Supervisor.
FR-6.2	Set price	High	allow the Admin to set the price for an approved stage, upon which the stage status shall become Price Proposed and the assigned worker shall be notified.
FR-6.3	Accept price	High	allow the assigned worker to Accept the proposed price, upon which the stage status shall become Price Accepted and then Completed.
FR-6.4	Decline price	High	allow the assigned worker to Decline the proposed price, upon which the stage status shall become Price Declined and the Admin shall be notified.
FR-6.5	Revise price	High	allow the Admin to set a revised price for a declined stage, returning it to Price Proposed for the worker to review.
FR-6.6	Pricing history	High	record every proposed price, revised price, acceptance, and decline with the actor, value, optional reason, and timestamp as an immutable pricing history.

Business Rules

Rule ID	Rule statement
BR-6.1	A stage price shall be set only after Supervisor approval of that stage.
BR-6.2	Only the worker assigned to a stage shall accept or decline that stage's price.
BR-6.3	Only an Admin shall set or revise a stage price.
BR-6.4	The price at Price Accepted shall be the amount used for that worker's payment for that stage.

3.2.7 Notifications

Keeps every party informed at each handoff so work never stalls waiting on a phone call.

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
A workflow event occurs (assignment, ready, inspection result, price proposed/revised, accept/decline)	The system sends an in-app notification to the party concerned
A mobile user has an event addressed to them	The system delivers a push notification where the device permits

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-7.1	Event alerts	High	send an in-app notification to the relevant recipient on each of: stage assignment, ready-for-inspection, inspection approval, inspection rejection, price proposed or revised, price acceptance, and price decline.

Req. ID	Title	Priority	The system shall …
FR-7.2	Unread indicator	Low	display an unread-notification indicator to each user until notifications are viewed.
FR-7.3	Push delivery	Medium	deliver notifications to mobile users via push notification where the device permits.

Business Rules

Rule ID	Rule statement
BR-7.1	A notification shall be addressed only to the role(s) party to the event that generated it.

3.2.8 Payments & Financial Records

Records what each worker has earned per completed stage and tracks whether it has been paid. An earning arises only once the worker has accepted the price set after Supervisor approval. The system records payment status; it does not move money.

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
A stage reaches Price Accepted (Completed)	The system creates an earning entry for the worker equal to the accepted price
An Admin marks an earning entry as Paid	The system records the payment date and acting Admin

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-8.1	Record earning	High	record, for each stage that reaches Price Accepted, an earning entry for the assigned worker equal to the accepted price.
FR-8.2	Payment status	High	maintain a payment status of Unpaid or Paid for each worker earning entry.
FR-8.3	Mark paid	Medium	allow an Admin to mark a worker earning entry as Paid, recording the date and the acting Admin.
FR-8.4	Guard earnings	High	prevent an earning entry from being created for a stage whose price has not been Accepted by the worker.

Business Rules

Rule ID	Rule statement
BR-8.1	A worker shall earn payment for a stage only after that worker has Accepted the stage's price, which follows Supervisor approval.
BR-8.2	Only an Admin shall change a payment status to Paid.

3.2.9 Dashboard & Reporting

Gives the Admin a live operational picture and gives each worker a transparent monthly statement of jobs and pay.

Stimulus / Response Sequence

Stimulus (user action / trigger)	System response
An Admin opens the dashboard	The system displays active-project counts, stage status counts, stages awaiting pricing, declined-price stages, and unpaid totals
A worker selects a month on their report	The system lists that month's completed jobs, accepted prices, and payment status
A worker taps 'Download PDF'	The system generates a PDF of the monthly report

Functional Requirements

Req. ID	Title	Priority	The system shall …
FR-9.1	Admin dashboard	High	present the Admin a dashboard showing counts of active projects, job-card stages by status, stages awaiting a price, stages with a declined price, and total unpaid payments.
FR-9.2	Action queues	High	present the Admin actionable queues of approved stages awaiting a price and declined-price stages awaiting a revised price.
FR-9.3	Monthly report	High	generate, for each Carpenter and Painter, a monthly report listing jobs completed, accepted price per job, total earned, amount paid, and amount outstanding for a selected month.
FR-9.4	PDF export	Medium	allow a worker to export their monthly report as a PDF.
FR-9.5	Workload view	Medium	display worker workload (assigned, in-progress, and overdue stages) to the Admin.

Business Rules

Rule ID	Rule statement
BR-9.1	A worker shall view monthly reports only for their own account.
BR-9.2	The monthly report shall include only stages Completed (price accepted) within the selected calendar month.

3.3 Usability Requirements

ID	Requirement	Verification
NFR-U.1	A worker shall be able to accept or decline a stage price within 3 taps from the mobile home screen.	Tap-count test
NFR-U.2	A first-time Supervisor shall complete an inspection (open job, upload photo, approve) without written instructions within 3 minutes.	Moderated usability test
NFR-U.3	Error messages shall state what went wrong and how to correct it.	QA checklist of error states

ID	Requirement	Verification
NFR-U.4	All interactive elements shall meet the minimum touch-target size defined by the target platform guidelines.	Accessibility scan
NFR-U.5	The Admin dashboard's primary KPIs shall be visible without scrolling on a 1366×768 display.	Inspection

3.4 Performance Requirements

ID	Area	Measurable Target
NFR-P.1	Screen load	Any primary screen (dashboard, job list, job detail) shall fully render within 3 seconds at p95 under normal load on a 4G connection.
NFR-P.2	Action response	A price set/accept/decline action shall complete within 2 seconds at p95 under normal load.
NFR-P.3	Concurrent users	The system shall support 200 simultaneous authenticated users without response-time degradation exceeding 20%.
NFR-P.4	Availability	The system shall achieve 99.5% monthly uptime excluding approved maintenance windows.
NFR-P.5	Photo upload	Upload of up to 10 inspection photos (max 5 MB each) shall complete within 15 seconds at p95 on 4G.
NFR-P.6	Report generation	A worker monthly report shall be generated within 5 seconds at p95.

3.5 Logical Database Requirements

The database design shall satisfy the following requirements-level entity model.

Entity	Key attributes	Relationships
User	id, name, contact, role, status	Assigned to Stages; acts on Payments
Project	id, name, client, deadline, status	Has many Job Cards
Job Card	id, project_id, title, description	Belongs to Project; has many Stages; has Attachments
Stage	id, job_card_id, type, assignee_id, price, status, timestamps	Belongs to Job Card; has Pricing History, Photos, Earning
Pricing History	id, stage_id, actor_id, value, action (proposed/revised/accepted/declined), reason, timestamp	Belongs to Stage; append-only; populated after approval
Inspection Photo	id, stage_id, supervisor_id, file_ref, timestamp	Belongs to Stage
Payment / Earning	id, stage_id, worker_id, amount, status, paid_date	Belongs to Stage and Worker
Notification	id, recipient_id, event_type, ref, read_flag, timestamp	Belongs to User

Data integrity rules:

- Every Project, Job Card, Stage, and Payment shall have a system-generated unique identifier.
- The Pricing History table shall be append-only; existing entries shall never be modified or deleted.

- Projects and Users shall be soft-deleted (archived/deactivated), never hard-deleted, where related records exist.
- Inspection photos and earning records shall be retained for a minimum defined period (see TBD-04).
- An Earning record shall exist only for a Stage whose price has been Accepted (status Completed).

3.6 Design Constraints

Category	Constraint
Branding	The application UI shall follow the DesignArc brand identity: forest green (#4E7A54) primary, brick red (#B23A2E) accent, cream (#F7F4EE) background.
Technology	Cross-platform mobile app and responsive web admin over a shared REST API and relational database, cloud-hosted (specific stack pending TBD-01).
Security	bcrypt password hashing (cost ≥12); TLS 1.2+ in transit; signed JWT session tokens; RBAC on every endpoint; access-controlled media storage.
Compliance	Support correction and erasure of personal data where no legal retention requirement applies; follow OWASP ASVS practices.

3.7 Standards Compliance

Standard	What compliance requires
ISO/IEC/IEEE 29148:2018	This SRS's structure and requirement quality criteria (this document is the evidence)
WCAG 2.1 AA	Colour contrast, touch-target sizing, and keyboard/screen-reader accessibility for key journeys
OWASP ASVS	Authentication, access-control, and data-protection verification during security testing
ISO 8601	All stored and displayed timestamps use the ISO 8601 date/time format

3.8 Software System Attributes

3.8.1 Reliability

The system shall guarantee that pricing history, inspection photos, and earning records are never lost once committed, and shall roll back any transaction that fails part-way so that no partial stage or payment record is created.

3.8.2 Availability

The system shall achieve 99.5% monthly uptime excluding approved maintenance windows, with automated health checks and alerting (see NFR-P.4).

3.8.3 Security

The system shall authenticate every request with a signed token, authorise every action through RBAC, hash passwords with bcrypt (cost ≥12), encrypt all traffic with TLS 1.2+, restrict media access to authorised users, and maintain an audit log of authentication failures and permission-denied events.

3.8.4 Maintainability

The system shall be built with a modular architecture, documented REST API, agreed coding standards, and automated tests achieving a minimum of 80% coverage on business-logic modules, delivered through a CI/CD pipeline.

3.8.5 Portability & Scalability

The mobile app shall run on current supported Android and iOS versions and the web app on current Chrome, Edge, Firefox, and Safari. The backend shall be deployable as containers and shall scale horizontally to meet the 200-concurrent-user target in NFR-P.3.

4 Verification

Every requirement is mapped to one of four verification methods — Test (T), Demonstration (D), Inspection (I), or Analysis (A). A requirement is considered verified only when evidence is recorded against its ID and reviewed by the responsible QA lead or stakeholder.

Method	Symbol	When used
Test	T	Requirements with measurable numerical targets (e.g., p95 response time)
Demonstration	D	Requirements shown working end-to-end (e.g., the inspection workflow)
Inspection	I	Requirements verified by reviewing code, config, or documentation (e.g., bcrypt use)
Analysis	A	Requirements proven by calculation or modelling (e.g., concurrency capacity)

Verification Matrix

Requirement Group	Primary Method	Verification Approach
FR-1.x — Auth & Roles	Test / Inspection	RBAC tests, negative permission tests, credential-flow tests
FR-2.x — Projects	Test / Demonstration	CRUD tests, completion roll-up checks
FR-3.x — Job Cards & Stages	Test / Demonstration	Status-machine tests, stage-sequence gate tests
FR-4.x — Stage Assignment	Test / Demonstration	Assignment tests, role-match and no-price-at-assignment checks
FR-5.x — Execution & Inspection	Demonstration / Test	End-to-end inspection walkthroughs, photo-required tests
FR-6.x — Post-Inspection Pricing	Test / Demonstration	Price-after-approval gate, accept/decline/revise flow, pricing-history immutability
FR-7.x — Notifications	Demonstration	Event-to-recipient delivery checks
FR-8.x — Payments	Test	Earning-on-accept guards, paid/unpaid state tests
FR-9.x — Dashboard & Reports	Test / Demonstration	KPI accuracy tests, monthly report and PDF checks
NFR-U.x — Usability	Demonstration / Inspection	Usability sessions, tap-count walkthroughs, accessibility scans
NFR-P.x — Performance	Test / Analysis	Load tests, p95 timing evidence, capacity analysis
Security requirements	Inspection / Test	Code review, configuration audit, penetration test

5 Supporting Information

5.A Appendix A — Requirement Traceability Matrix (RTM)

The RTM is maintained as a live document and updated throughout development. It maps each requirement to its source, delivery version, verification method, and test case. Representative rows:

Requirement	Source	Version	Verification	Test Case
FR-1.2 — RBAC	Workshop [2]	MVP	Test	TC-101
FR-5.4 — Approve stage	Workshop [2]	MVP	Demo	TC-154
FR-6.3 — Accept price	Workshop [2]	MVP	Test / Demo	TC-163
FR-9.3 — Monthly report	Workshop [2]	v1.0	Test	TC-193
NFR-P.1 — Screen load	Perf objective	v1.0	Test / Analysis	TC-301

5.B Appendix B — Phased Feature Roadmap

Version	Focus	Representative Features
MVP	Core workflow value	Roles & auth, projects, job cards & stages, stage assignment, inspection with photos, post-inspection pricing (set → accept/decline), per-stage earnings, basic dashboard
v1.0	Full first release	Full KPI dashboard & approval queue, monthly PDF reports, push notifications, completion roll-up, workload views
v2.0	Expansion & automation	Separate Finance role, richer analytics, overdue/deadline alerts
v3.0	Advanced capabilities	Materials/inventory, client portal, accounting integration, forecasting

5.C Appendix C — TBD Resolution Log

TBD ID	Open item	Owner	Deadline
TBD-01	Confirm technology stack (mobile framework, hosting provider)	QueMore Tech Lead	Design
TBD-02	Confirm currency/locale handling (LKR only or multi-currency)	DesignArc	Design
TBD-03	Confirm whether a separate Finance role is required for payments	DesignArc	Sprint 1
TBD-04	Confirm retention period for inspection photos and records	DesignArc / Legal	Sprint 1
TBD-05	Confirm maximum photo count and size per inspection	QueMore / DesignArc	Design

6 References to Verification Artefacts

Verification artefacts are maintained as separate controlled documents and referenced here, keeping this SRS focused on requirements.

Artefact	Description	Owner
Requirement Traceability Matrix	Live RTM mapping every FR to test case and version	QueMore QA Lead
Test Specifications	Detailed test cases derived from FR, BR, and NFR IDs	QueMore QA Team
Usability Study Reports	Moderated and unmoderated usability session results	QueMore UX Lead
Performance Test Results	Load test reports with p95/p99 evidence	QueMore QA / DevOps
Penetration Test Report	Security vulnerability assessment results	Security Team

Final Approval

This SRS is baselined when all approvers have signed. A baselined SRS may only be changed through the formal change-control process, which requires a new version number, a revision-history entry, and re-approval by affected parties.

Role	Name	Signature / Date
Business Owner (DesignArc)		
Product Owner (QueMore)		
Development Lead (QueMore)		
QA Lead (QueMore)		

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